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2.1.1 Low rank approximation

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columns in
this case. So we can't
do that. Now there's
something called the
pseudo-inverse that
takes its place. And
the pseudo-inverse is
given by $A^T (A A^T)^{-1} A^T b_j$.
And you can prove
that this is the best
choice in the case
where A has linearly
independent
columns. Alright, and
what does
that mean? Well that
means that our b_j is
approximately A
times that vector
which is given by that
right there. Now what

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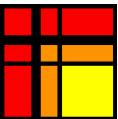
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Reading assignment

1/1 point (graded)



Read Unit 2.1.1 of the notes. [\[LINK\]](#)

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Homework 2.1.1.1

1/1 point (graded)
If B is $m \times n$ and A is $m \times k$, how many entries are there in A and X ? (For context, see [Section 2.1.1 of the notes.](#))

Entries in A :

☐ mn

☒ mk

☐ kn



Entries in X :

☐ mn

☐ mk

☒ kn



For full solution see notes: [[LINK](#)]

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Answers are displayed within the problem

Homework 2.1.1.2

1/1 point (graded)
A X is called a rank- k approximation of B . Why? (For context, see [Section 2.1.1 of the notes.](#))

☒ The matrix AX has a rank of at most k

☐ The matrix AX has a rank of exactly k

☐ The matrix AX has a rank of at least k



Now prove it!

For full solution see notes: [[LINK](#)]

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